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Studierichting: Informatica
Oefeningenreeks 4A nr. 63.3

$$\int \sqrt{(1-x^2)^3}dx$$

Neem:

$x = \cos t$

$dx = -\sin t dt$

Dan:

$$\int \sqrt{(1-x^2)^3}dx = -\int \sqrt{(1-\cos^2 t)^3} \sin t dt$$
$$= -\int \sqrt{\sin^6 t} \sin t dt$$
$$= -\int \sin^3 t \sin t dt$$
$$= -\int \sin^4 t dt$$
$$= -\int \left(\frac{1-\cos 2t}{2}\right)^2 dt$$
$$= -\frac{1}{4}\int (1 - 2\cos 2t + \cos^2 2t)dt$$
$$= -\frac{1}{4}\int 1dt + \frac{1}{2}\int \cos 2t dt - \frac{1}{4}\int \cos^2 2t dt$$
$$= -\frac{1}{4}t + \frac{1}{4}\sin 2t - \frac{1}{4}\int \frac{1+\cos 4t}{2}dt$$
$$= -\frac{1}{4}t + \frac{1}{4}\sin 2t - \frac{1}{8}\int (1+\cos 4t)dt$$
$$= -\frac{1}{4}t + \frac{1}{4}\sin 2t - \frac{1}{8}t - \frac{1}{32}\sin 4t + c$$
$$= -\frac{3}{8}t + \frac{1}{4}\sin 2t - \frac{1}{32}\sin 4t + c$$

Terug substitueren:

$t = \mathrm{Bgcos}\, x$

$$= -\frac{3}{8}\mathrm{Bgcos}\, x + \frac{1}{4}\sin(2\,\mathrm{Bgcos}\, x) - \frac{1}{32}\sin(4\,\mathrm{Bgcos}\, x) + c$$

References